Nature Exposure Report

Kimble County, TX | 2,340 acres | TX-KIM-2024-04817

Prepared for	AXA XL Sustainability
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Basal Informatics

Ground-truth ecological data for nature-risk assessment

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Executive Summary

CRITICAL

76
Feral Hog Exposure Score

This assessment covers a 2,340-acre parcel in Kimble County, TX. Camera trap monitoring detected 12 species across the property, including 2 invasive species (Feral Hog, Axis Deer). The overall nature exposure risk is rated **CRITICAL** with a Feral Hog Exposure Score of 76/100. Estimated annual invasive species damage totals $114,533. Detection frequencies are IPW-adjusted for non-random camera placement.

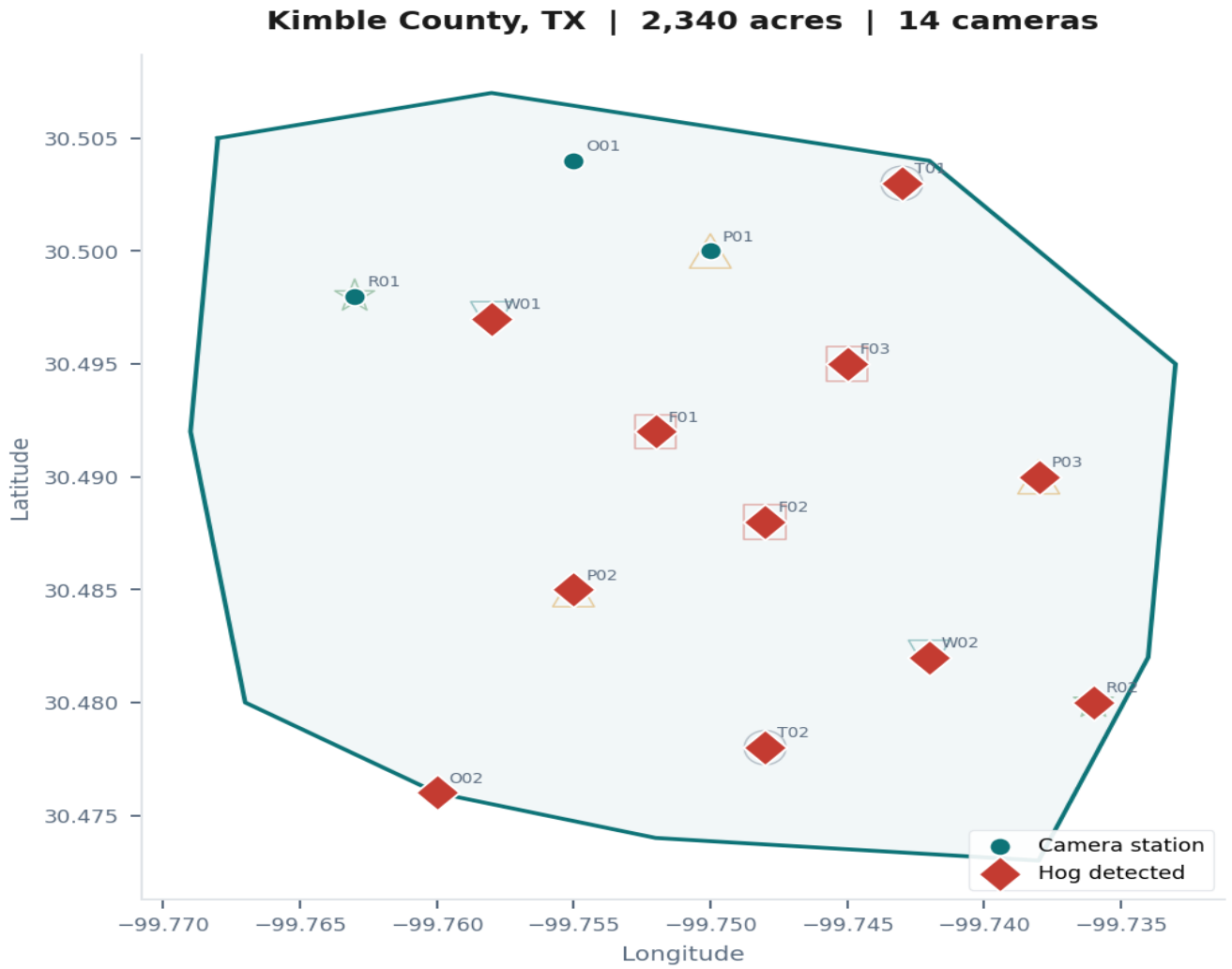
Key Findings

- **Feral hog damage estimated at $107,835/year** (10-yr NPV: $832,670), based on IPW-adjusted detection frequency of 63.4% across 2,340 acres.
- **1 ESA-listed species** with potential habitat overlap identified. Estimated compliance cost: $15,000–$85,037.
- **Bias correction reduced feral hog detection frequency by 15.2 percentage points** (raw 78.6% → adjusted 63.4%), correcting for non-random camera placement near feeders and water sources.
- **Monitoring gaps identified** requiring an estimated 17 additional cameras to improve corridor coverage. Current data confidence: grade F.

Assessment Metrics

Species detected	12	**Invasive species**	2
Est. annual loss	$114,533	**ESA species flagged**	1
Monitoring months	11	**Camera density**	1.11/km²
Data confidence	F	**Bias correction**	Applied

Parcel Overview



Camera stations shown as colored markers. Marker size indicates detection volume. Red markers highlight feral hog detections. Parcel boundary shown in teal. Dashed lines indicate habitat unit boundaries.

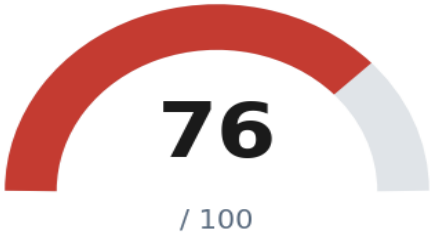
Species Inventory

Species	Events	Det. Freq.	Conf.	Risk Flag
Feral Hog	674	63.4%	D	INVASIVE — MODERATE
Axis Deer	55	33.3%	D	INVASIVE — MODERATE
White-tailed Deer	1384	100.0%	C-	—
Raccoon	378	93.8%	F	—
Nine-banded Armadillo	230	72.0%	F	—
Wild Turkey	442	66.0%	D	—
Coyote	189	65.6%	F	—
Eastern Cottontail	79	61.9%	F	—
Bobcat	110	46.1%	D	—
Virginia Opossum	40	26.1%	F	—
Gray Fox	10	23.9%	F	—
Red Fox	20	16.6%	F	—

Detection frequencies are IPW-adjusted for non-random camera placement (Kolowski & Forrester 2017). Raw frequencies before correction are available in the full data export.

Events = independent detections (30-min threshold). Confidence grades reflect corridor coverage, temporal completeness, and detection frequency (A = highest).

Invasive Species Damage Projection

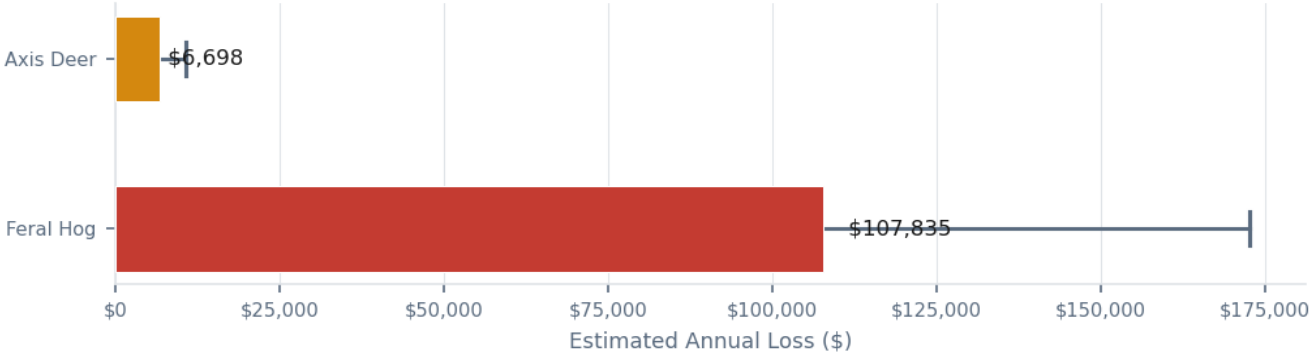


$107,835
Estimated Annual Loss

$832,670 10-Year NPV

Confidence interval (±60%): $43,134 – $172,535

Damage Estimates with Confidence Intervals



Model Assumptions

Feral Hog

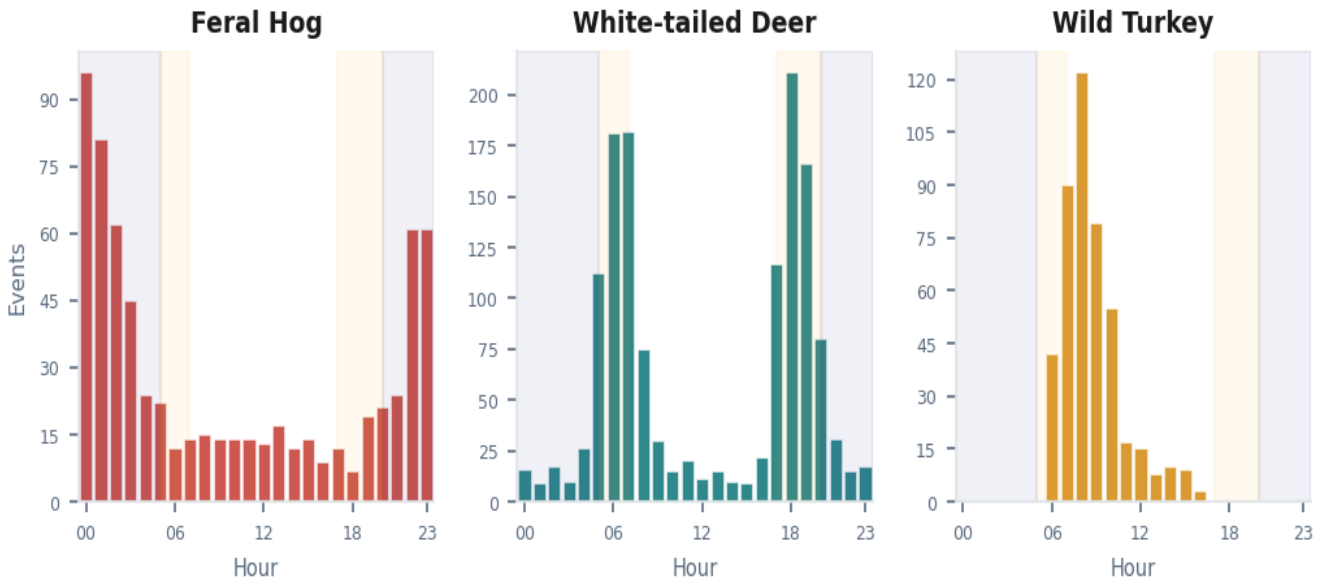
Parameter	Value
Base rate source	USDA-APHIS base rates (USDA-APHIS)
Base cost	$53.79/acre/year
Ecoregion calibration	1.15×
Detection frequency (IPW-adjusted)	63.4%
Logistic frequency scale	0.7450
Parcel acreage	2,340 acres
Discount rate (NPV)	5.0%
NPV horizon	10 years
Confidence grade	D
CI width	±60%

Axis Deer

Parameter	Value
Base rate source	USDA-APHIS base rates (Texas Parks & Wildlife estimate)
Base cost	$12.50/acre/year
Ecoregion calibration	1.10×
Detection frequency (IPW-adjusted)	33.3%
Logistic frequency scale	0.2082
Parcel acreage	2,340 acres
Discount rate (NPV)	5.0%
NPV horizon	10 years
Confidence grade	D
CI width	±60%

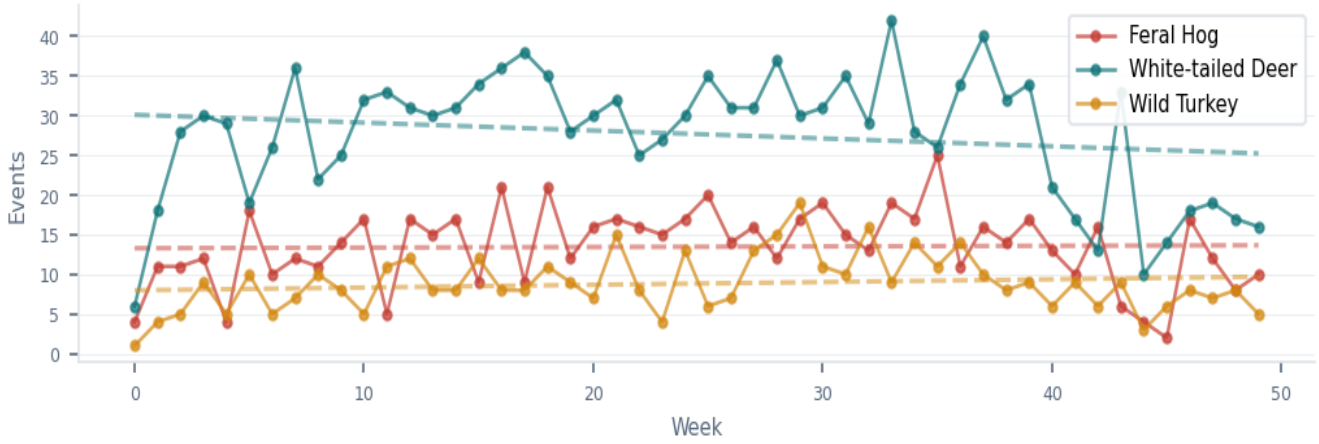
Detection frequency is a relative activity index, not absolute density. Broadley et al. 2020 showed density-dependent movement can cause cameras to underestimate population declines by up to 30%.

Activity Patterns



Independent events by hour of day. Shading indicates dawn (05:00–07:00), dusk (17:00–20:00), and night periods.

Weekly Activity Trend



Weekly independent event counts. Trend lines show direction of activity over the monitoring period.

Feral Hog: Nocturnal (66% nocturnal, 21% diurnal)

White-tailed Deer: Crepuscular (10% nocturnal, 36% diurnal)

Wild Turkey: Diurnal (0% nocturnal, 90% diurnal)

Data Confidence

Overall confidence grade	F
Monitoring duration	11 months
Camera density	1.11 cameras/km²
Bias correction	Applied

Species Confidence Grades

Species	Det. Freq.	Grade	Cameras	Events
Feral Hog	63.4%	D	11/14	674
Axis Deer	33.3%	D	4/14	55
White-tailed Deer	100.0%	C-	14/14	1384
Raccoon	93.8%	F	12/14	378
Nine-banded Armadillo	72.0%	F	8/14	230
Wild Turkey	66.0%	D	9/14	442
Coyote	65.6%	F	10/14	189
Eastern Cottontail	61.9%	F	7/14	79

+ 4 additional species (see Species Inventory table for full list)

Regional Classification Accuracy

Source: **hunter-verified corrections** | Validation status: **literature_baseline**

Species	Accuracy
White Tailed Deer	99.1%
Turkey	96.6%
Feral Hog	94.2%

Detection-to-density calibrated via paired surveys at Matagorda Bay. Classification accuracy from hunter corrections in Edwards Plateau habitat units.

Top Monitoring Gaps

Gaps in corridor coverage where additional cameras would most improve data confidence.

Corridor	Habitat Unit	Gap (m)	Species Affected	Cameras Needed
Forest Grass Edge	HU-1209020104-30a-41	1487	Cottontail Rabbit	7
Forest Grass Edge	HU-1209020104-30a-41	1436	Cottontail Rabbit	7

Riparian	HU-1209020104-30a-41	1271	Feral Hog	3

Methodology

Detection Pipeline

Camera trap images are processed through a two-stage classification pipeline. Stage 1 uses Microsoft MegaDetector v5 to locate animals in each frame (minimum confidence threshold: 0.3). Stage 2 applies Google SpeciesNet for taxonomic classification of detected animals. Both models are pre-trained on large-scale camera trap datasets and deployed without site-specific fine-tuning to maintain generalization across deployment regions.

Event Independence

Raw detections are grouped into independent events using a two-threshold system. First, photos within 60 seconds of each other from the same camera and species are grouped as a single trigger burst. Second, bursts separated by fewer than 30 minutes are consolidated into one independent event. This 30-minute independence threshold is standard in camera trap ecology and prevents repeated counting of the same individual. All metrics in this report use independent event counts, never raw photo counts.

Confidence Calibration

Raw classifier confidence scores are calibrated using temperature scaling (Dussert et al. 2025), which applies a learned temperature parameter (T = 1.08) to soften overconfident predictions by 5–10%. Calibrated probabilities more accurately reflect true classification accuracy. Binary softmax entropy is computed for each prediction; detections exceeding the entropy threshold (0.59 nats) are flagged for human review. Temporal priors (Mac Aodha et al. 2019) further refine predictions by incorporating species-specific circadian activity patterns.

Placement Bias Correction

Camera traps are typically placed near feeders, water sources, trails, and food plots — locations chosen to maximize game detection. Kolowski & Forrester (2017) demonstrated that trail and feeder cameras detect 9.7× more animals than randomly placed cameras. This placement bias inflates naive detection rates.

We correct for this bias using inverse probability weighting (IPW) adapted from the causal inference literature (Robins, Hernán & Brumback 2000). A logistic regression propensity model estimates P(camera placed here | landscape covariates) for each camera location relative to 500 uniformly sampled reference points within the parcel boundary. Covariates include distance to water, distance to road, slope, canopy cover, relative elevation, distance to habitat edge, land cover class, and aspect. Placement context labels (feeder, trail, etc.) are excluded from the model to prevent trivial separation.

Cameras in biased locations (near feeders, water) receive lower weights; cameras in landscape-representative locations receive higher weights. Weights are stabilized by the marginal probability of camera placement and capped at an 8:1 max-to-min ratio to prevent extreme influence from any single camera. This approach follows the eBird framework (Johnston et al. 2021) for correcting spatial sampling bias in citizen science biodiversity data.

Regional Accuracy Validation

Classification accuracy cannot be reliably estimated from out-of-region validation data (Beery, MIT CSAIL). Actual accuracy depends on local species composition, habitat structure, and camera placement — factors that vary between deployments. Our regional accuracy metrics are derived from hunter-verified corrections within each deployment region: hunters review flagged detections and correct misclassifications, providing ground-truth labels specific to the monitored landscape.

Limitations

Detection frequency is a relative abundance index, not absolute population density. Conversion to absolute density requires site-specific calibration. Parsons et al. (2017) demonstrated $R^2 = 0.80$ correlation between camera trap relative abundance indices and true density under standardized protocols. However, Broadley et al. (2020) identified that density-dependent movement patterns can cause cameras to underestimate population declines by up to 30% — as populations decline, remaining animals expand their home ranges, maintaining camera encounter rates despite reduced abundance. Our Matagorda Bay calibration program addresses this through paired camera trap and field survey validation.

Damage projections are model-based estimates, not observed losses. Actual damage depends on land use, management practices, crop type, and seasonal patterns not fully captured by detection frequency alone. Confidence intervals are provided to reflect uncertainty proportional to data quality.

References

Broadley, K. et al. (2020). Density-dependent space use affects interpretation of camera trap detection rates. Ecology and Evolution, 9(24), 14031-14041.

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Kolowski, J.M. & Forrester, T.D. (2017). Camera trap placement and the potential for bias due to trails and other features. PLOS ONE, 12(10), e0186679.

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Parsons, A.W. et al. (2017). Mammal communities are larger and more diverse in moderately developed areas. eLife, 7, e38012.

Robins, J.M., Hernán, M.A. & Brumback, B. (2000). Marginal structural models and causal inference in epidemiology. Epidemiology, 11(5), 550-560.

Basal Informatics

Ground-truth ecological data for nature-risk assessment. We deploy scalable camera trap networks across private land, process imagery through calibrated AI classifiers, and deliver bias-corrected species inventories and damage projections to insurers and lenders for TNFD and EU CSRD compliance.

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